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Onset of primary moult in immature and breeding adult Common Swifts *Apus apus*

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ABSTRACT

We studied the primary moult of immature and adult breeding Common Swift *Apus apus* in Italy. Birds were sampled at breeding colonies, or by attracting them to mist nets with the playback of recorded calls. Our sample of 590 immatures (2cy) assessed that about a quarter (25.6%) of these birds start moulting the inner primaries during their stay in Europe. We find that this moult starts usually in the second half of June, earlier than was observed in other European countries, where individuals begin moulting in July. Interestingly, and in contrast to previous knowledge on the moulting schedules of breeding Common Swifts, we found that more than half of breeding adults (56%, 15 of 27) started their primary moult in July. Moult before the swifts left our study area was limited to the first two inner primaries. Our study showed that breeding Commons Swifts are able to regulate their moult when already engaged in rearing young, and both immature and adult Swifts start their moult before leaving Europe towards their African winter quarters.

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Moult is an important physiological process in birds and one of the most demanding of energy (Newton 2009). The pattern of feather replacement can differ widely between different species, to the point where ornithological terminology includes several terms to describe moult (ascendant, descendant, complete, partial, suspended, multiple, etc; Campbell & Lack 2011). Additionally, different moult patterns are widely reported between age groups and subspecies of the same species, as well as variations in the starting date of moult across different populations (Jenni & Winkler 1994, Newton 2009).

The energetically challenging periods of moulting and breeding are usually temporally separated in temperate birds, but can occur simultaneously in some taxa. An overlap between the renewal of primaries and reproduction is predicted to be the optimal strategy only for birds living in habitats with low seasonality, for example in the tropics (Barta *et al* 2006, McNamara & Houston 2008). In nature, however, it also has been observed in extreme northern breeders (Hemborg *et al* 1998). In these birds, which stay only a few weeks in the breeding quarters, the overlap between reproduction and primary moult is thought to be an adaptation to time constraints, or a consequence of them.

As generally reported among swifts, the northern species have a pattern of separated moult and reproduction, possibly as an adaptation to the short time spent on the breeding grounds during their single annual breeding attempt. Southern and tropical swifts have moult and breeding seasons that overlap broadly, while in other species of the northern hemisphere the overlap is only partial (Marín & Stiles 1993, Bull & Collins 1993, Marín 2003, Garcia-Del-Rey *et al* 2008, Collins 2010).

In Europe, an enlightening example is the comparison of Common Swift *Apus apus* and Pallid Swift *A. pallidus*. The Common Swift lays a single clutch and is generally reported to undertake a complete moult in its winter quarters in Africa (de Roo 1966, Weitnauer 1977, Glutz von Blotzheim & Bauer 1980, Cramp 1985, Baker 2016), while the Pallid Swift moults five or six inner primaries in Europe during the rearing period of their second brood (Boano *et al* 2015).

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Recently, Jukema et al (2015) showed that 23% of second-calendar-year immature Common Swifts in the Netherlands moult at least one innermost primary in July, while none of the adult Swifts were in active primary moult in the region. This percentage is similar to the value given for the same species in July–August before migration in Dorset, UK (20%, Ginn & Melville 1983): these authors attribute their sample to adult birds, however. A peculiar case of moult in Europe was reported by Kolunen & Peiponen (1991), who found 57% of adult Common Swifts moulting in Finland in 1986. In this instance, the birds had delayed their autumn migration due to unexpected adverse climatic conditions in the breeding area, and the period of moult in Europe overlapped with the normal period of moult in the African wintering quarters.

With this paper, we add information on the primary moult of the Common Swift in Italy, in the southern part of the European breeding range. In particular, we investigate whether only immatures engaged in moult or whether breeding adults were involved as well.

Methods

Breeding adults (n = 27) were sampled in July at five colonies located in northern Italy: Carignano, 44.91°N 7.68°E; Carmagnola, 44.85°N 7.72°E; Fossano, 44.55°N 7.72°E, Piemonte; Jerago con Orago, 45.71°N 8.79°E, Lombardia; and Castellaro di Guiglia, 44.39°N 10.95° E, Emilia Romagna. Adults were sampled when rearing young, either by catching birds by hand in the nest cavity, at sites where nests were located on buildings with cavities inspectable from the inside, or by trapping them with a hand net placed in front of the entrance hole. In the Jerago colony, catching started from 1 July 2021, while in all the other colonies only a few birds were still engaged in breeding activities after mid-July.

The moult of non-breeding birds was examined using a sample of 680 birds mist netted between May and early July near Castelfidardo (43.46°N 13.55°E, Marche) by one of the authors (PM). All these individuals were mist netted away from breeding colonies by attracting them with the playback of a screaming call (Malacarne *et al* 1989). According to the state of primary wear, the sample was composed mainly of immature birds in their second calendar year (2cy), n = 590 or after second calendar year (>2cy), n = 90. These older birds, all attracted with playback during 9 May to 3 July, were presumed also to be non-breeding birds, as one might guess from their typical 'bangers' behaviour (Lack & Lack 1952, Genton 2010); their age could be ascertained only for six individuals, all of them known to be in their third calendar year because they had been ringed as 2cy the previous year.

Moult was recorded using the method described by Ginn & Melville (1983) in which each flight feather is given a score from 0 (old) to 5 (fully grown and new); primaries were numbered descendantly, from the innermost (P1) to the outermost (P10). Primary moult score is summed across the primaries and has a maximum value of 50. In this study, we deal exclusively with primary moult. We did not collect data on the moult of feathers on other parts of the body, with the exception of the tail which was never found to be moulting.

Results

We did not observe any immature Common Swifts starting to moult their primaries in May (Table 1). In June, from 17.2 to 32.5% of 2cy immatures were found moulting in different years (26.6% overall), and in July all birds sampled had started their moult. Moulting immatures began replacing primaries 1 and 2 (Figure 1), with the majority of moulting birds only just having initiated primary moult and with only the innermost primary feather growing (total moult score 1), though the most advanced individuals had a moult score of 11 (Figure 2). By contrast, only one of the 90 > 2cy birds mist netted near Castelfidardo had begun to moult.

With similar timing, the first breeding adult Common Swifts starting their moult were found in the breeding colony of Jerago at the beginning of July (Figure 3). Birds captured after the middle of July were found to be replacing primaries 1 and 2, reaching the maximum recorded score of 10 from the last third of July (Table 2; Figure 2).

Table 1. Incidence of moult among 590 immature Swifts (2cy, EURING age 5) captured near Castelfidardo, Marche, during 2019–21.

Year	Month	n	Number moulting	% moulting
By year and month				
2019	June	259	79	30.5%
	July	17	17	100%
2020	May	3	0	0%
	June	180	31	17.2%
	July	3	3	100%
2021	May	11	0	0%
	June	117	38	32.5%
By month				
	May	14	0	0%
	June	556	148	26.6%
	July	20	20	100%
Total		590	168	28.4%



Figure 1. Immature (2cy) Common Swift captured on 26 June 2016, growing two inner primaries and their primary coverts. Photo by P. Micheloni.

Discussion

In this study, we found that non-breeding 2cy immature Common Swifts began to moult one to three inner primaries during their short summer stays in Europe. This is in line with the findings of Jukema *et al* (2015) in the Netherlands. The percentage of immatures starting their moult in the Netherlands (23%) is similar to the value observed in Italy (26.6%) in June. In Italy, however, we observed an earlier onset of moult dates than Jukema *et al* (2015) found, with a moult start of P1 (innermost primary) as early as 10 June and a progression such that in July 100% of the individuals were in moult, compared with 60% in the second half of July in the Netherlands. The state of moult allows individuals to sometimes be identified as 2cy even when observed in flight or at least when examining good-quality photographs of flying birds (Figure 4).

Contrary to what is commonly reported in the literature (Cramp 1985), adult Common Swifts can start their primary moult during the breeding period;

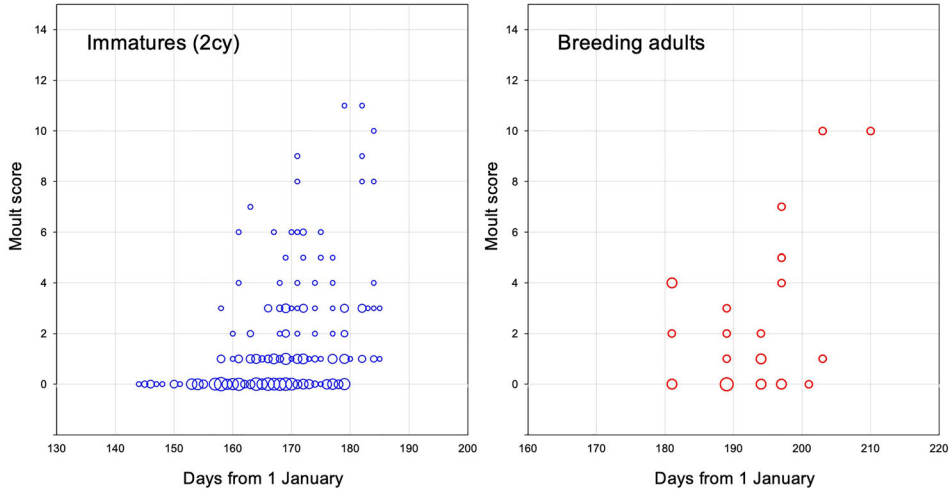


Figure 2. Primary moult scores of moulting 2cy and breeding adult Common Swifts in northern Italy, plotted against date (days from 1 January). Symbol size indicates numbers of individual birds (blue dots 1–44 birds, red dots 1–5 birds).

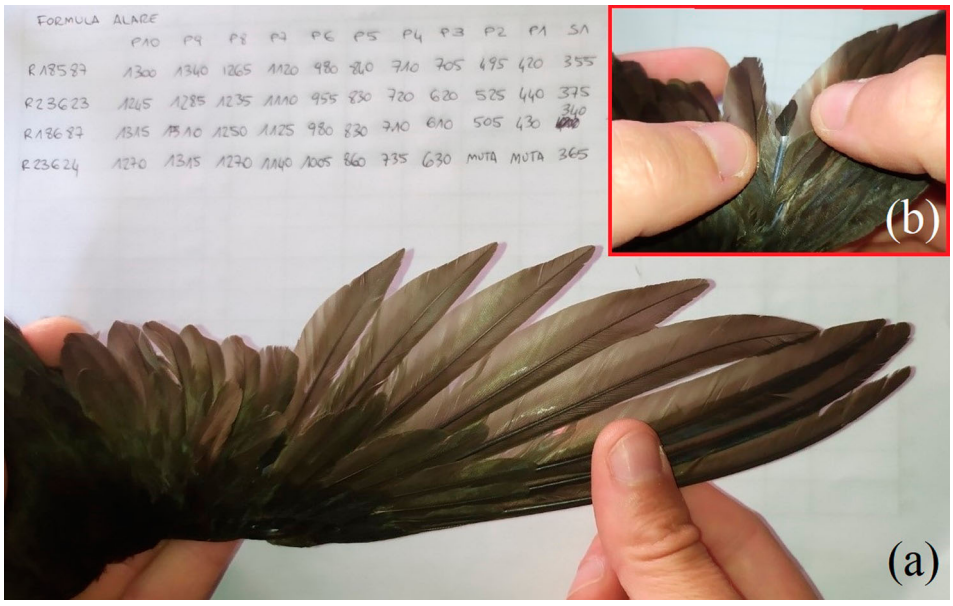


Figure 3. Adult Common Swifts moulting while breeding at Jerago, Province of Varese, on 9 July 2021: (a) left wing of the ringed bird R23624 with growing P1 and P2 hidden by secondaries; (b) growing P1 of the ringed bird R23625 (see Table 2). Photo by M. Manica.

these breeding birds can start moulting in the first days of July, though, as far as is known, the moult does not involve more than two inner primaries (P1 and P2) before the birds leave their breeding quarters. From the start of July, therefore, individuals that are moulting can include both immatures (2cy) and breeding adults (Figure 5). Interestingly, these observations do not differ much from the moult timing of the Pallid Swift *Apus pallidus* observed in the same northern Italian area. In adult breeding Pallid Swifts, indeed, the primary moult only rarely starts before the middle of July and primary score

Table 2. Primary moult scores of 27 breeding adult Common Swifts (>2cy) from five northern Italian localities in the provinces of Cuneo, Turin, Modena and Varese. Site codes: CN1 Fossano (Piemonte); TO1 Carmagnola; TO2 Carignano (Piemonte); MO1 Castellaro (Emilia-Romagna); VA1 Jerago con Orago (Lombardia). Primaries are numbered descendantly, from innermost P1 to outermost P10.

Site	Ring no.	Date	Nest contents	P1	P2	P3–P10	Moult score	First encounter
CN1	R03215	23 Jul 2021	unknown	1	0	0	1	newly ringed
TO2	R03224	17 Jul 2018	unknown	5	2	0	7	newly ringed
TO2	R03225	17 Jul 2018	unknown	3	1	0	4	newly ringed
TO2	R03226	30 Jul 2018	unknown	5	5	0	10	newly ringed
TO1	R02029	1 Jul 2021	unknown	2	0	0	2	newly ringed
TO1	R02030	1 Jul 2021	unknown	4	0	0	4	newly ringed
TO1	R02031	1 Jul 2021	unknown	3	1	0	4	newly ringed
TO1	R02032	1 Jul 2021	unknown	0	0	0	0	newly ringed
TO1	R02033	1 Jul 2021	unknown	0	0	0	0	newly ringed
MO1	W115806	17 Jul 2021	pulli	4	1	0	5	2019, breeding adult
MO1	W101434	17 Jul 2021	pulli	0	0	0	0	2013, nestling (9cy)
MO1	R09157	17 Jul 2021	pulli	0	0	0	0	2014, nestling (8cy)
VA1	R18587	9 Jul 2021	2 pulli	0	0	0	0	2019, breeding adult
VA1	R18607	9 Jul 2021	3 pulli	0	0	0	0	2019, breeding adult
VA1	R18684	9 Jul 2021	3 pulli	0	0	0	0	2019, breeding adult
VA1	R18685	9 Jul 2021	2 pulli	1	0	0	1	2020, breeding adult
VA1	R18687	9 Jul 2021	3 pulli	0	0	0	0	2020, breeding adult
VA1	R23623	9 Jul 2021	pulli	0	0	0	0	2021, breeding adult
VA1	R23624	9 Jul 2021	1 pullus	2	1	0	3	2021, breeding adult
VA1	R23625	9 Jul 2021	1 pullus	2	0	0	2	2021, breeding adult
VA1	R18560	14 Jul 2021	2 pulli	0	0	0	0	2018, breeding adult
VA1	R18572	14 Jul 2021	2 pulli	1	0	0	1	2019, non-breeding adult
VA1	R18587	14 Jul 2021	2 pulli	0	0	0	0	2019, breeding adult
VA1	R18689	14 Jul 2021	1 pullus	2	0	0	2	2020, breeding adult
VA1	R23626	14 Jul 2021	2 pulli	1	0	0	1	2021, breeding adult
VA1	R18542	21 Jul 2021	2 pulli	5	5	0	10	2018, breeding adult
VA1	R18632	21 Jul 2021	3 pulli	0	0	0	0	2019, breeding adult



Figure 4. This Common Swift, with an advanced moult involving two inner primaries growing, photographed near Fossano, Province of Cuneo, on 25 June 2017, is surely a subadult (2cy) as shown also by the wear of the pointed inner primaries P3–P5 and the pointed outer rectrices. Photo by M. Giordano.



Figure 5. This moulting Common Swift photographed on 9 July 2019 near Fossano, Province of Cuneo, P1 having been dropped from right wing and on left wing just dropping, cannot be safely aged by its moult stage alone. It appears to be an adult, judging by the rounded tip of the longest primary P9 and by the relatively fresh inner primaries P2–P5. Photo by M. Giordano.

does not exceed 11 by the end of July (Boano *et al* 2015, GB unpublished).

We did not collect data on the moult of other parts of the body, except for the tail, which was never found in moult. Future studies could clarify whether the moult process in Europe also can involve other body feathers, or if it is limited to the first few primaries.

Our study shows that breeding Common Swifts can initiate their moult while still engaged in rearing young. This process could be problematic, with possible conflicts between the hormones that regulate gonadal development and the behaviours linked to parental care and those regulating moult. Also, the activities related to parental duties and feather replacement are both energetically demanding and must be traded off (Drent & Daan 1980, Buttemer *et al* 2020).

Moreover, the differences apparent between the Netherlands and Italian Common Swift populations in the seasonal timing of the onset of moult suggest that it could be interesting to investigate ecological correlates across a larger sample of locations. In particular, considering the results of Pérez-Granados *et al* (2021) about the effect of time-constraints on the extent of post-juvenile moult in a genus of long-distance migrant birds, it could be interesting to compare different species and populations of swifts showing different migration strategies and distances (Norevik *et al* 2019, Åkesson *et al* 2020). At the same time, it would be useful to evaluate the effects of the peak abundance of insect prey (Bryant 1975, Cucco & Malacarne 1996, Hallmann *et al* 2017) because it could influence the availability of food in an especially energy-demanding period (Barta *et al* 2008).

In conclusion, we show that both immature and adult Common Swifts, from a population breeding in the south of the European range, start their primary moult before leaving Europe for their African winter quarters.

Moreover, we wish to highlight the usefulness of the playback method using screaming calls in attracting subadult swifts in late spring and early summer. This technique could be crucial for future studies aimed to examine the start of the moult in other areas located both at the northern or the southern boundaries of the Common Swift breeding range. We recommend researchers, in particular, to carefully check the moult of breeding adults, especially from mid-July, when inspecting occupied nests.

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References

Åkesson, S., Atkinson, P.W., Bermejo, A., de la Puente, J., Ferri, M., Hewson, C.M., Holmgren, J., Kaiser, E., Kearsley, L., Klaassen, R.H.G., Kolunen, H., Matsson, G., Minelli, F., Norevik, G., Pietiäinen, H., Singh, N.J., Spina, F., Viktora, L. & Hedenström, A. (2020) Evolution of chain migration in an aerial insectivorous bird, the common swift *Apus apus*. *Evolution* 74, 2377–2391.

Baker, J. (2016) *Identification of European Non-Passerines*. Second edition. British Trust for Ornithology, Thetford.

Barta, Z., Houston, A.I., McNamara, J.M., Welham, R.K., Hedenström, A., Weber, T.P. & Feró, O. (2006) Annual routines of non-migratory birds: optimal moult strategies. *Oikos* **112**, 580–593.

Barta, Z., McNamara, J.M., Houston, A.I., Weber, T.P., Hedenström, A. & Feró, O. (2008) Optimal moult strategies in migratory birds. *Philosophical Transactions of the Royal Society B: Biological Sciences* **363**, 211–229.

Boano, G., Pellegrino, I. & Cucco, M. (2015) Moult and morphometrics of the pallid swift *Apus pallidus* in northwestern Italy. *Ardeola* **62**, 35–48.

Bryant, D.M. (1975) Breeding biology of House Martins *Delichon urbica* in relation to aerial insect abundance. *Ibis* **117**, 180–216.

Bull, E.L. & Collins, C.T. (1993) Nesting chronology, molt and ectoparasites of Vaux's Swifts in northeastern Oregon. *Avocetta* **17**, 203–207.

Buttemer, W.A., Addison, B.A. & Klasing, K.C. (2020) The energy cost of feather replacement is not intrinsically inefficient. *Canadian Journal of Zoology* **98**, 142–148.

Campbell, B. & Lack, E. (eds) (2011) *A dictionary of birds*. T & AD Poyser, London.

Collins, C.T. (2010) Notes on the breeding biology of the white-throated swift in southern California. *Bulletin, Southern California Academy of Sciences* **109**, 23–36.

Cramp, S. (1985) *Handbook of the birds of Europe, the Middle East and North Africa: the birds of the Western Palearctic*. Volume IV. Terns to Woodpeckers. Oxford University Press, Oxford.

Cucco, M. & Malacarne, G. (1996) Reproduction of the pallid swift (*Apus pallidus*) in relation to weather and aerial insect abundance. *Italian Journal of Zoology* **63**, 247–253.

De Roo, A. (1966) Age-characteristics in adult and subadult swifts, *Apus a. apus* (L.), based on interrupted and delayed wing-moult. *Le Gerfaut* **56**, 113–134.

Drent, R.H. & Daan, S. (1980) The prudent parent: energetic adjustments in avian breeding. *Ardea* **68**, 225–252.

Garcia-Del-Rey, E., Gosler, A.G., Gonzalez, J. & Wink, M. (2008) Sexual size dimorphism and moult in the Plain Swift *Apus unicolor*. *Ringing & Migration* **24**, 81–87.

Genton, B. (2010) Chronologie comportementale du Martinet noir *Apus apus* sur un site de reproduction. Choisir une cavité, la partager, la défendre. *Nos Oiseaux* **57**, 243–264.

Ginn, H.B. & Melville, D.S. (1983) *Moult in birds*. BTO Guide 19. British Trust for Ornithology, Tring.

Glutz von Blotzheim, U.N. & Bauer, K.M. (1980) *Handbuch der Vogel Mitteleuropas*. Volume 9. Akademische Verlagsgesellschaft, Wiesbaden.

Hallmann, C.A., Sorg, M., Jongejans, E., Siepel, H., Hofland, N., Schwan, H., Stenmans, W., Müller, A., Sumser, H., Hörren, T., Goulson, D. & de Kroon, H. (2017) More than 75 percent decline over 27 years in total flying insect biomass in protected areas. *PLoS ONE* **12**, e0185809.

Hemborg, C., Lundberg, A. & Siikamäki, P. (1998) Trade-off between reproduction and moult – a comparison of three Fennoscandian pied flycatcher populations. *Oecologia* **117**, 374–380

Jenni, L. & Winkler, R. (1994) *Moult and Ageing of European Passerines*. Academic Press, London.

Jukema, J., van de Wetering, H. & Klaassen, R.H.G. (2015) Primary moult in non-breeding second-calendar-year Swifts *Apus apus* during summer in Europe. *Ringing & Migration* **30**, 1–6.

Lack, D. & Lack, E. (1952) The breeding behaviour of the Swift. *British Birds* **45**, 186–215.

Kolunen, H. & Peiponen, V.A. (1991) Delayed autumn migration of the Swift *Apus apus* from Finland in 1986. *Ornis Fennica* **68**, 81–92.

Malacarne, G., Palomba, I., Griffa, M., Castellano, S. & Cucco, M. (1989) Quantitative analysis of differences in the vocalizations of the Common Swift *Apus apus* and the Pallid Swift *Apus pallidus*. *Avocetta* **13**, 9–14.

Marín, M. (2003) Molt, plumage, body mass, and morphometric of a population of White-throated Swift in southern California. *Western Birds* **34**, 216–224.

Marín, M. & Stiles, F.G. (1993) Notes on the biology of the Spot-fronted Swift. *Condor* **95**, 479–483.

McNamara, J.M. & Houston, A.I. (2008) Optimal annual routines: behaviour in the context of physiology and ecology. *Philosophical Transactions of the Royal Society B: Biological Sciences* **363**, 301–319.

Newton, I. (2009) Moult and plumage. *Ringing & Migration* **24**, 220–226.

Norevik, G., Boano, G., Hedenström, A., Lardelli, R., Liechti, F. & Åkesson, S. (2019) Highly mobile insectivorous swifts perform multiple intra-tropical migrations to exploit an asynchronous African phenology. *Oikos* **128**, 640–648.

Pérez-Granados, C., Calleja, J.A., Seoane, J., Fernández-Tizón, M., Vigalondo, B., Meléndez, L. & Kiat, Y. (2021) Time available for moulting shapes inter- and intra-specific variability in post-juvenile moult extent in wheatears (genus *Oenanthe*). *Journal of Ornithology*, **162**, 255–264.

Weitnauer, E. (1977) Von der Mauser der Handschwingen beim Mauersegler *Apus apus*. *Ornithologische Beobachter* **74**, 89–94.